BIOLOGY COLLOQUIUM SERIES | FALL 2026

Applications of Quantitative Ecology to Imperiled Ecosystem Management

Mon, October 26th
12:00 – 1:30 PM
RB 130

Dr. Chris Searcy

University of Miami / College of Arts and Sciences

In this talk, I focus on the three systems with long-term data. First, I explore South Florida's iconic habitat, the Everglades, which is the site of the world's largest restoration project. The restoration program has focused on hydrological management to improve the quality and quantity of freshwater reaching the still intact portions of the historic Everglades. At the same time, invasive species have emerged as a leading threat to the Everglades and South Florida more broadly. I will present our literature review examining the reasons why invasive species are such a major problem in South Florida, our field data demonstrating the severity of the threat posed by the invasive Asian swamp eel, and our experimental evidence that hydrological management can be an effective tool to control South Florida invasives. Second, I share our investigations into the conservation of two federally endangered salamanders in coastal California. We found healthier populations for both species when multiple breeding ponds are in close vicinity. This argues for the importance of metapopulation dynamics in amphibian conservation. Third, I visit the Florida Scrub, which has the highest rate of endemism in the southeastern US. Our previous work has shown that one of the endangered endemics in this habitat is dependent on soil microbes for its persistence. In this talk, I demonstrate the utility of ecological niche modelling to identify which aspects of the soil microbiome are most important for governing plant distributions, and which members of the plant community are most responsive to microbial effects.